

QCE25 Technical Papers
Albuquerque Convention Center
August 31 – September 5, 2025

v77

QCE25 Best Technical Papers			
Track	1st	2nd	3rd
QALG	425	271	282
QAPP	725	178	551
QML	153	269	635
QNET	492	226	353
QPHO	395	337	836
QSYS	311	327	523
QTEM	689	631	123
Overall QCE25 Best Technical Paper			
IEEE Quantum Technical Community (QTC)			
Distinguished Best Technical Paper Award			
QAPP 725			

Search for ECID

TP02::QML_ — Trainable and Optimized Quantum Neural Networks for Machine Learning
 Sunday, 31 August, 15:00 – 16:30

260 — Differentiable Quantum Architecture Search in Quantum-Enhanced Neural Network Parameter Generation — Samuel Yen-Chi Chen, Chen-Yu Liu, Kuan-Cheng Chen, Wei-Jia Huang, Yen-Jui Chang and Wei-Hao Huang

564 — Trainable Quantum Neural Network for Multiclass Image Classification with the Power of Pre-trained Tree Tensor Networks — Keisuke Murota and Takumi Kobori

119 — Learning to Learn with Quantum Optimization via Quantum Neural Networks — Kuan-Cheng Chen, Hiromichi Matsuyama and Wei-Hao Huang

TP03::QAPP — Efficient Simulation of Physical Systems via Quantum Circuit Design
 Sunday, 31 August, 15:00 – 16:30

584 — Scalable quantum circuits of exponential of Pauli-strings and its applications to Hamiltonian simulations — Rohit Sarma Sarkar, Sabyasachi Chakraborty and Bibhas Adhikari

398 — Hamiltonian Simulation for solving the Advection Equation with arbitrary velocity fields — Niladri Gomes, Gautam Sharma and Jay Pathak

551 — Efficient Classical-Processing of Constant-Depth Time Evolution Circuits in Control Hardware — Akhil Francis, Abhi Rajagopala, Norm M. Tubman, Katherine Klymko and Kasra Nowrouzi

TP04::QAPP — Financial Foresight with Quantum Tools

Sunday, 31 August, 15:00 – 16:30

597 — LEP-QNN: Loan Eligibility Prediction Using Quantum Neural Networks — Nouhaila Innan, Alberto Marchisio, Mohamed Bennai and Muhammad Shafique

532 — Introducing the Quantum Economic Advantage Online Calculator — Frederick Mejia, Hans Gundlach, Jayson Lynch, Carl Dukatz, Andrew Lucas, Eleanor Crane, Prashant Shukla and Neil Thompson

304 — Adaptive Non-local Observable on Quantum Neural Networks — Hsin-Yi Lin, Huan-Hsin Tseng, Samuel Yen-Chi Chen and Shinjae Yoo

TP05::QSYS — Metrics for Quantum Runtime and Performance

Sunday, 31 August, 15:00 – 16:30

432 — Is Circuit Depth Accurate for Comparing Quantum Circuit Runtimes? — Matthew Tremba, Ji Liu and Paul Hovland

430 — A Practical Framework for Assessing the Performance of Observable Estimation in Quantum Simulation — Siyuan Niu, Efehan Kökcü, Sonika Johri, Anurag Ramesh, Avimita Chatterjee, David Bernal Neira, Daan Camps and Thomas Lubinski

608 — Quantum Processing Unit (QPU) processing time Prediction with Machine Learning — Lucy Xing, Sanjay Kumar Lalta Prasad Vishwakarma, David Kremer, Francisco Martín-Fernández, Ismael Faro and Juan Cruz-Benito

TP06::QSYS — Calibration and Testing for Real-World Quantum Devices

Sunday, 31 August, 15:00 – 16:30

96 — Tailored Quantum Device Calibration with Statistical Model Checking — Filip Mazurek, Marissa D'Onofrio, Andrew Van Horn, Jiyong Yu, Kavyashree Ranawat, Jungsang Kim and Kenneth Brown

227 — From Ideal to Noisy: Adapting Property-Based Testing for Real-World Noisy Quantum Computers — Gabriel Pontolillo, Asmar Muqeet, Shaukat Ali and Mohammad Reza Mousavi

405 — Towards a Digital Twin of Noisy Quantum Computers: Calibration-Driven Emulation of Transmon Qubits — Ronny Mueller, Maximilian Zanner, Mika Schielein, Martin Rüfenacht, Elise Jennings and Cica Gustiani

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TP07::QALG — Scalable and Adaptive QAOA for Real-World Optimization I

Sunday, 31 August, 10:00 – 11:30

228 — Quantum tree generator improves QAOA state-of-the-art for the knapsack problem — Paul Johann Christiansen, Lennart Binkowski, Debora Ramacciotti and Sören Wilkening

801 — Evolving a multi-population evolutionary-QAOA on distributed QPUs — Francesca Schiavello, Edoardo Altamura, Ivano Tavernelli, Stefano Mensa and Benjamin Symons

TP08::QALG — Scalable and Adaptive QAOA for Real-World Optimization II**Sunday, 31 August, 13:00 – 14:30**

681 — A Noise-Aware Scalable Subspace Classical Optimizer for Quantum Approximate Optimization Algorithm — Kwassi Joseph Dzahini, Jeffrey Larson, Matt Menickelly and Stefan Wild

796 — Combinatorial Optimization with Few Qubits — Rudy Raymond, Alexander Buts and Marco Pistoia

TP09::QALG — Formulations and Frameworks for Constrained Combinatorial Optimization**Sunday, 31 August, 15:00 – 16:30**

784 — QUBO formulations for some graph-theory problems: Snake-in-the-box and Coil-in-the-box — Federico Fuidio, Eduardo Canale and Rafael Sotelo

438 — Standardization of Multi-Objective QUBOs — Loong Kuan Lee, Thore Gerlach and Nico Piatkowski

811 — SCOOP: A Quantum-Computing Framework for Constrained Combinatorial Optimization — Prashanti Priya Angara, Emily Martins, Ulrike Stege and Hausi A Muller

TP10::QAPP — Quantum Optimization for Real-World Scheduling and Logistics**Sunday, 31 August, 10:00 – 11:30**

355 — Application-Driven Benchmarking of the Traveling Salesperson Problem: a Quantum Hardware Deep-Dive — Amine Bentellis, Benedikt Poggel and Jeanette Lorenz

749 — A New Hybrid Algorithm for Solving the Unit Commitment Problem — Willie Aboumrad, Phani R V Marthi, Suman Debnath, Martin Roetteler and Evgeny Epifanovsky

449 — Solving a real-world modular logistic scheduling problem with a quantum-classical metaheuristics — Florian Krellner, Abhishek Awasthi, Nico Kraus, Sarah Braun, Michael Poppel and Daniel Porawski

TP11::QAPP — Quantum Methods for Power and Physical System Simulation**Sunday, 31 August, 13:00 – 14:30**

404 — A Quantum-Inspired Algorithm for Wave Simulation Using Tensor Networks — Kevin Lively, Vittorio Pagni and Gonzalo Camacho

527 — Quantum Computing-Enabled Contingency Analysis for Power Systems — Yuqi Jiang, Zhiding Liang, Yan Li, Qiang Guan and Ganesh Kumar Venayagamoorthy

742 — Noise-aware Quantum Annealing for State Estimation in Power Systems — Quoc-Bao Tran and Thang Dinh

TP12::QAPP — Variational Quantum Imaginary Time Evolution for Optimization and Simulation**Sunday, 31 August, 15:00 – 16:30**

253 — Solving Constrained Combinatorial Optimization Problems with Variational Quantum Imaginary Time Evolution — Xin Wei Lee and Hoong Chuin Lau

313 — Accelerating Large-Scale Linear Algebra Using Variational Quantum Imaginary Time Evolution — Willie Aboumrad, Daiwei Zhu, Claudio Grotto, François-Henry Rouet, Jezer Jojo, Robert Lucas, Jay Pathak, Ananth Kaushik and Martin Roetteler

357 — Avoiding convergence stagnation in a quantum circuit evolutionary framework through an adaptive cost function — Bruno Oziel Fernandez, Rodrigo Bloot and Marcelo Moret

TP13::QNET — Quantum Repeaters**Sunday, 31 August, 15:00 – 16:30**

186 — A Comparative Study of All-photonic and Memory-based Quantum Repeaters and Networks — Chuen Hei Chan, Charu Jain, Ezra Kissel, Wenji Wu, Edwin Barnes, Sophia E. Economou and Inder Monga

188 — Integrating Entanglement Purification into All-Photonic Quantum Repeaters — Naphan Benchasattabuse, Michal Hajdušek and Rodney Van Meter

248 — Quantum repeaters enhanced by vacuum beam guides — Yu Gan, Mohadeseh Azari, Nitish Chandra, Xin Jin, Jinglei Cheng, Kaushik P. Seshadreesan and Junyu Liu

TP14::QNET — Post-Quantum Cryptography**Sunday, 31 August, 15:00 – 16:30**

422 — A Survey of Post-Quantum Cryptography Support in Cryptographic Libraries — Nadeem Ahmed, Lei Zhang and Aryya Gangopadhyay

542 — Evaluating Post-Quantum Cryptographic Algorithms on Resource-Constrained Devices — Jesus Lopez, Viviana Cadena and Mohammad Saidur Rahman

TP15::QSYS — Quantum Circuit Cutting**Sunday, 31 August, 15:00 – 16:30**

279 — Quantum Circuit Cutting: A Security Methodology — George Typaldos, Theodoros Trochatos and Jakub Szefer

762 — Is Circuit Cutting Scalable for Practical Quantum Applications? — Songqinghao Yang and Prakash Murali

420 — A Formalization of Measurement-Commuting Unitaries — Ulrik de Muelenaere, Sinan Pehlivanoglu, Amr Sabry and Peter Kogge

TP16::QSYS — Quantum Intermediate Representation**Sunday, 31 August, 15:00 – 16:30**

453 — Towards a pulse-level intermediate representation for diverse quantum control systems — Jude Alnas, Aniket S. Dalvi and Kenneth R. Brown

149 — QIRopt: An Optimization Method for Quantum Intermediate Representation — Junjie Luo, Haoyu Zhang and Jianjun Zhao

136 — Bifrost: Topology-Aware Approximate Compilation of Variational Quantum Algorithms — Vivek Yanamadula, Jonathan Yang and Runzhou Tao

TP17::QAPP — Quantum and Quantum-Inspired Annealing for Scalable Optimization I**Sunday, 31 August, 15:00 – 16:30**

377 — Quantum Annealing Hyperparameter Analysis for Optimal Sensor Placement in Production Environments — Nico Kraus, Marvin Erdmann, Alexander Kuzmany, Daniel Porawski and Jonas Stein

693 — Vector Annealing, a quantum-inspired technique: Benchmarking Performance Against Quantum and Simulated Annealing within the BACQ Framework — Stephane Louise

346 — Towards solving large QUBO problems using quantum algorithms: improving the 'LogQ' scheme — Yagnik Chatterjee and Jérémie Messud

TP18::QAPP — Compilation, Correlations, and Entanglement for Application-Driven Design**Sunday, 31 August, 15:00 – 16:30**

537 — Compilation of QCrank Encoding Algorithm for a Dynamically Programmable Qubit Array Processor — Jan Balewski, Wan-Hsuan Lin, Anupam Mitra, Milan Kornjaca, Stefan Ostermann, Pedro Lopes, Daniel Bochen Tan and Jason Cong

443 — A deep dive into the interplay of structured quantum peaked circuits and infinite temperature correlation functions — Myeongsu Kim, Manas Sajjan and Sabre Kais

609 — Stochastic Entanglement Configuration for Constructive Entanglement Topologies in Quantum Machine Learning with Application to Cardiac MRI — Mehri Mehrnia and Mohammed S.M. Elbaz

TP19::QAPP — Quantum Approaches to Simulation and Forecasting**Monday, 1 September, 15:00 – 16:30**

521 — Variational Quantum Linear Solver for Simulating Quantum Transport in Nanoscale Semiconductor Devices — Qimao Yang and Jing Guo

400 — Quantum Reservoir Computing for Corrosion Prediction in Aerospace: A Hybrid Approach for Enhanced Material Degradation Forecasting — Akshat Tandon, James Brown, Kenny Heitritter, Tarini Hardikar, Kanav Setia, Jasper Simon Krauser, Rene Boettcher and Klaus Schertler

725 — Quantum-Enhanced Parameter-Efficient Learning for Typhoon Trajectory Forecasting — Chen-Yu Liu, Kuan-Cheng Chen, Yi-Chien Chen, Samuel Yen-Chi Chen, Wei-Hao Huang, Wei-Jia Huang and Yen-Jui Chang

TP20::QSYS — Compiling for Fault Tolerant Quantum Computing

Monday, 1 September, 15:00 – 16:30

655 — A Lazy Resynthesis Approach for Simultaneous T Gate and Two-Qubit Gate Optimization of Quantum Circuits — Mu-Te Lau, Hsiang-Chun Yang, Hsin-Yu Chen and Chung-Yang Ric Huang

320 — Lattice Surgery Compilation Beyond the Surface Code — Laura S. Herzog, Lucas Berent, Aleksander Kubica and Robert Wille

168 — Making Neural Networks More Suitable for Approximate Clifford+T Circuit Synthesis — Mathias Weiden, Justin Kalloor, John Kubiawicz and Costin Iancu

TP21::QSYS — Specialized Quantum Workflows

Monday, 1 September, 15:00 – 16:30

219 — A New Perspective of Constructing Resource-Efficient Data-Lookup Quantum Oracles — Mingfei Yu, Mathias Soeken and Giovanni De Micheli

245 — SimQFL: A Quantum Federated Learning Simulator with Real-Time Visualization — Ratun Rahman, Atit Pokharel, Md Raihan Uddin and Dinh Nguyen

370 — HiPARS: Highly-Parallel Atom Rearrangement Sequence — Jonas Winklmann and Martin Schulz

TP22::QNET — Secure and Scalable Quantum Communication

Monday, 1 September, 10:00 – 11:30

200 — New Key Rate Bound for High-Dimensional BB84 with Multiple Basis Measurements — Trevor Thomas and Walter Krawec

573 — Time-Aware Qubit Assignment and Circuit Optimization for Distributed Quantum Computing — Leo Sünkel, Jonas Stein, Maximilian Zorn, Thomas Gabor and Claudia Linnhoff-Popien

556 — Servicing Demands between Multiple Pairs of Nodes on Heterogeneous Quantum Networks — Anoop Kumar Pandey, Bheemarjuna Reddy Tamma and Panduranga Rao Mv

TP23::QNET — Loss-Resilient Protocols for Quantum Communication

Monday, 1 September, 13:00 – 14:30

309 — A Quantum Speedup in Localizing Transmission Loss Change in Optical Networks — Yufei Zheng, Yu-Zhen Janice Chen, Prithwish Basu and Don Towsley

734 — Power Reduction of SFQ Readout Circuit for Superconducting Quantum Computers — Masumi Kanda, Yuki Matsumoto, Teruo Tanimoto, Masamitsu Tanaka and Takatsugu Ono

356 — Performance Analysis of MDI-QKD in Thermal-Loss and Phase Noise Channels — Heyang Peng, Seid Koudia, Leonardo Oleynik and Symeon Chatzinotas

TP24::QNET — Quantum Communication

Monday, 1 September, 15:00 – 16:30

285 — On the impact of classical and quantum communication networks upon modular quantum computing architecture system performance — Pau Escofet, Abhijit Das, Sahar Ben Rached, Santiago Rodrigo, Jordi Domingo, Fabio Sebastiano, Masoud Babaie, Batuhan Keskin, Edoardo Charbon, Peter Haring Bolívar, Maurizio Palesi, Elena

468 — Temperature dependence of a warm ensemble based memory on quantum communication rates — Kenny Gregory, Khaled Mnaymneh, Connor Kupchak and Andrew Macrae

570 — Lower Bounds for the Communication Costs of Distributing Quantum Fourier Transform on Clique Networks — Ali Ebneenasir

TP25::QSYS — Symbolic Tools and Resource Optimization for Quantum Architectures

Monday, 1 September, 15:00 – 16:30

229 — Symbolic Hamiltonian Compiler for Hybrid Qubit-Boson Processors — Ethan Decker, Evan McKinney, Erik Gustafson, Lucas Goetz, Alex Jones, Ang Li, Alexander Schuckert, Samuel Stein, Gushu Li and Eleanor Crane

403 — Optimal Qubit Reuse for Quantum Computation with Gate Reordering and Eager Measurement — Wen-Chen Yu, Tian-Fu Chen, Yu-Hsiang Chan, Jie-Hong Roland Jiang, Dah-Wei Chiou and Yao-Wen Chang

522 — Optimal Fermion-Qubit Mappings via Quadratic Assignment — Mitchell Chiew, Cameron Ibrahim, Sergii Strelchuk and Ilya Safran

TP26::QML_ — Noise Mitigation and Error Correction in QML

Monday, 1 September, 15:00 – 16:30

316 — Differentially Private Federated Quantum Learning via Quantum Noise — Atit Pokharel, Ratun Rahman, Shaba Altaf Shaon, Thomas Morris and Dinh C. Nguyen

343 — Constructing Optimal Noise Channels for Enhanced Robustness in Quantum Machine Learning — David Winderl, Nicola Franco and Jeanette Miriam Lorenz

635 — A Privacy-Preserving Federated Framework with Hybrid Quantum-Enhanced Learning for Financial Fraud Detection — Abhishek Sawaika, Swetang Krishna, Tushar Tomar, Durga Pritam Suggiseti, Aditi Lal, Tanmaya Shrivastav, Nouhaila Innan and Muhammad Shafique

TP27::QAPP — Toward Trustworthy Quantum Computation

Tuesday, 2 September, 15:00 – 16:30

505 — ABS-Q: An Effective Analysis-by-Synthesis Framework for Recovering Signals from Noisy Quantum Outputs — Phat Thanh Tran, Jake Scally, Austin Myers, Ryan Carmichael, Hyunjin Yi, Xiuwen Liu and Bayaner Arigong

612 — Enhancing Fidelity in NISQ Computers via Crosstalk-Optimized Gate Decomposition — Indu Palanisamy, Hieu Nguyen, Ethan Hunt, Dillon Horton, Dheeraj Tukkugudam and Tu Nguyen

618 — Statistical Assertions for Debugging Quantum Circuits and States in CUDA-Q — Jocelyn Li, Ella Rubinshtein and Margaret Martonosi

TP28::QAPP — Quantum Optimization for Logistics and Routing in Real-World Systems

Tuesday, 2 September, 15:00 – 16:30

181 — Optimizing Package Delivery with Quantum Annealers: Addressing Time-Windows and Simultaneous Pickup and Delivery — Eneko Osaba, Esther Villar-Rodriguez, Pablo Miranda-Rodriguez and Antón Asla

414 — QUADRO: A Hybrid Quantum Optimization Framework for Drone Delivery — James Holliday, Darren Blount, Hoang Nguyen, Khoa Luu and Samee Khan

563 — Path Matters: Industrial Data Meet Quantum Optimization — Lukas Schmidbauer, Carlos A. Riofrio, Florian Heinrich, Vanessa Junk, Ulrich Schwenk, Thomas Husslein and Wolfgang Maurer

TP29::QAPP — Quantum and Quantum-Inspired Annealing for Scalable Optimization II

Tuesday, 2 September, 10:00 – 11:30

254 — Implementing Slack-Free Custom Penalty Function for QUBO on Gate-Based Quantum Computers — Xin Wei Lee and Hoong Chuin Lau

711 — Solving Large-Scale QUBO with Transferred Parameters from Multilevel QAOA of low depth — Bao Bach, Filip Maciejewski and Ilya Saфро

450 — Quantum Boltzmann Machines using Parallel Annealing for Medical Image Classification — Danielle Schuman, Mark V. Seebode, Tobias Rohe, Maximilian Balthasar Mansky, Michael Schroedl-Baumann, Jonas Stein, Claudia Linnhoff-Popien and Florian Krellner

TP30::QAPP — Quantum Solutions for Transportation, Routing, and Scheduling Problems

Tuesday, 2 September, 13:00 – 14:30

669 — Solving Large-Scale Vehicle Routing Problems with Hybrid Quantum-Classical Decomposition — Andrew Maciejunes, John Stenger, Daniel Gunlycke and Nikos Chrisochoides

360 — QUEST: QUAntum-Enhanced Shared Transportation — Chinonso Onah, Neel Miscasci, Carsten Othmer and Kristel Michielsens

605 — Iterative quantum annealing for job shop scheduling with a directed graph representation — Rebecca Casati, Thomas Giudici, Pietro Torta and Enrico Prati

TP31::QAPP — Toward Robust and Secure Quantum Algorithms and Compilation

Tuesday, 2 September, 15:00 – 16:30

602 — Modal analysis on quantum computers via qubitization — Yasunori Lee and Keita Kanno

506 — RDQKL: Robust Decentralized Quantum Kernel Learning for Noisy and Adversarial Environment — Wenxuan Ma, Kuan-Cheng Chen, Shang Yu, Mengxiang Liu and Ruilong Deng

130 — QSeer: A Quantum-Inspired Graph Neural Network for Parameter Initialization in Quantum Approximate Optimization Algorithm Circuits — Lei Jiang, Chi Zhang and Fan Chen

TP32::QNET — Optimization in Quantum Networks

Tuesday, 2 September, 10:00 – 11:30

488 — Enhancing Quantum Federated Learning with Fisher Information-Based Optimization — Amandeep Singh Bhatia and Sabre Kais

600 — Realizing Scalability Limits of Quantum Communication Networks — Connor Howe and Ali Anwar

778 — A Hybrid Metaheuristic for Route Optimization in Quantum Repeater Networks — Suleiman Aliyu and Hongji Yang

TP33::QNET — Quantum Protocols and Experiments Beyond the Lab

Tuesday, 2 September, 13:00 – 14:30

250 — Evaluation of Distillation's Real-world Performance on a Superconducting Quantum Computer — Hikaru Yokomori, Marii Koyama, Naphan Benchasattabuse, Michal Hajdusek, Rodney Van Meter and Shota Nagayama

794 — How to share Multipartite Entanglement in a Real-World Linear Network Connecting Two Metropolises — Janka Memmen and Anna Pappa

289 — Boson Sampling from a Multimode Cavity Containing Bose-Einstein Condensate — Vitaly Kocharovsky

TP34::QNET — Real-Time Control & Integration for Scalable Quantum Platforms

Tuesday, 2 September, 15:00 – 16:30

237 — Orchestrating Multi-Zone Shuttling in Trapped-Ion Quantum Computers — Daniel Schoenberger and Robert Wille

769 — A Scalable Real-Time Decoder for Quantum Error Correction Based on Hyperdimensional Computing — Rob A. Damsteegt, Masoud Babaie, Sebastian Feld and Fabio Sebastiano

789 — Partitioning Cryogenic Integrated Electronics for Scalable Spin Qubit Operation — Lotte Geck, Pau Dietz Romero, Lammert Duipmans and Stefan van Waasen

TP35::QALG — Toward Secure and Reliable Quantum Computing in the NISQ Era

Tuesday, 2 September, 10:00 – 11:30

678 — Enhanced Local Differential Privacy Protection against Crosstalk Attacks in Quantum Computing — Hui Zhong, Xinyue Zhang, Hao Wang, Shucheng Yu, Yu Wang and Miao Pan

102 — Digital Zero-Noise Extrapolation with Quantum Circuit Unoptimization — Elijah Pelofske and Vincent Russo

484 — Quantum Public Key Encryption for NISQ Devices — Nishant Rodrigues, Walter Krawec, Brad Lackey, Deb Mukhopadhyay and Bing Wang

TP36::QALG — Overcoming Optimization Challenges in Quantum Algorithms

Tuesday, 2 September, 13:00 – 14:30

238 — From barren plateaus through fertile valleys: Conic extensions of parameterised quantum circuits — Lennart Binkowski, Gereon Koßmann, Tobias J. Osborne, René Schwonnek and Timo Ziegler

561 — Dynamic Range Reduction via Branch-and-Bound — Thore Gerlach and Nico Piatkowski

740 — Efficient Quantum Gradient and Higher-order Derivative Estimation via Generalized Hadamard Test — Dantong Li, Dikshant Dulal, Mykhailo Ogorodnikov, Hanrui Wang and Yongshan Ding

TP37::QALG — Structured and Fault-Tolerant Circuits for Robust Quantum Computation

Tuesday, 2 September, 15:00 – 16:30

496 — Structured Clifford+T Circuits for Efficient Generation of Quantum Chaos — Asim Sharma and Avah Banerjee

531 — Turbo-Annihilation of Hook Errors in Stabilizer Measurement Circuits — Michele Pacenti, Asit K. Pradhan, Shantom K. Borah and Bane Vasic

282 — Fault Tolerant Quantum Simulation via Symplectic Transvections — Zhuangzhuang Chen, Jack Weinberg and Narayanan Rengaswamy

TP38::QAPP — Quantum Algorithms for Learning and Combinatorial Optimization

Tuesday, 2 September, 10:00 – 11:30

637 — Cross-Problem Parameter Transfer in Quantum Approximate Optimization Algorithm: A Machine Learning Approach — Kien Nguyen, Bao Bach and Ilya Safro

365 — A quantum search method for quadratic and multidimensional knapsack problems — Sören Wilkening, Andreea-Iulia Lefterovici, Lennart Binkowski, Marlene Funck, Michael Perk, Robert Karimov, Sándor Fekete and Tobias J. Osborne

264 — Bayesian Quantum Orthogonal Neural Networks for Anomaly Detection — Natansh Mathur, Brian Coyle, Nishant Jain, Snehal Raj, Akshat Tandon, Jasper Simon Krauser and Rainer Stoessel

TP39::QAPP — Quantum Computing for Financial Modeling and Portfolio Optimization
Tuesday, 2 September, 10:00 – 11:30

151 — Implementing Credit Risk Analysis with Quantum Singular Value Transformation — Davide Veronelli, Francesca Cibrario, Emanuele Dri, Valeria Zaffaroni, Giacomo Ranieri, Davide Corbelleto and Bartolomeo Montrucchio

225 — Higher-Order Portfolio Optimization with Quantum Approximate Optimization Algorithm — Valter Uotila, Julia Ripatti and Bo Zhao

131 — Advancing Bi-objective Weighted Portfolio Optimization Across International Markets via Quantum Annealing — Shu-Yu Kuo and Yao-Hsin Chou

TP40::QAPP — Quantum Algorithms for Chemistry and Biological Systems I
Wednesday, 3 September, 10:00 – 11:30

185 — A Quantum Approach for Optimal Transient Control in Network-Based Epidemic Models — Deborah Volpe, Giacomo Orlandi, Mattia Boggio, Lorenzo Zino, Carlo Novara and Giovanna Turvani

209 — Identifying Protein Co-regulatory Network Logic by Solving B-SAT Problems through Gate-based Quantum Computing — Aspen Erlandsson Brisebois, Jason Broderick, Zahed Khatooni, Heather Wilson, Steven Rayan and Gordon Broderick

197 — Determining molecular ground state with quantum imaginary time evolution using broken-symmetry wave function — Pawan Sharma Poudel, Kenji Sugisaki, Michal Hajdušek and Rodney Van Meter

TP41::QAPP — Quantum Algorithms for Chemistry and Biological Systems II
Wednesday, 3 September, 13:00 – 14:30

533 — Quantum Advantage in Computational Chemistry? — Hans Gundlach, Keeper Sharkey, Jayson Lynch, Victoria Hazoglou, Carl Dukatz, Kung-Chuan Hsu, Eleanor Crane, Karin Walczyk, Marcin Bodziak and Neil Thompson

814 — Orbital-rotated Fermi-Hubbard model as a benchmarking problem for quantum chemistry with the exact solution — Ryota Kojima, Masahiko Kamoshita and Keita Kanno

625 — QGENE: A Quantum Dynamic Programming Model for Pairwise Sequence Alignment — Shyam Sundar, Hari Priya Narahari, Sadanand Venkataraman, Cherisha Ronney, Madhankumar P, Sharmila Devi V and Santhi Natarajan

TP42::QAPP — merging Topics in Quantum Applications
Wednesday, 3 September, 15:00 – 16:30

98 — A Quantum Algorithm Based Heuristic to Hide Sensitive Itemsets — Abhijeet Ghoshal, Yan Li, Syam Menon and Sumit Sarkar

336 — Sampling-based Quantum Optimization Algorithm with Quantum Relaxation — Hiromichi Matsuyama and Yu Yamashiro

479 — CHSH Violations using Dynamic Circuits — Jean-Baptiste Waring, Christophe Pere and Sébastien Le Beux

TP43::QALG — Quantum Algorithms for Linear Systems and Period Finding

Wednesday, 3 September, 10:00 – 11:30

362 — Depth-Based Matrix Classification for the HHL Quantum Algorithm — Mark Danza, Sonia Lopez Alarcon and Cory Merkel

381 — Practical Scalability of LuGo: Benchmarking the HHL Algorithm Using an Enhanced QPE Algorithm — Chao Lu, Murali Gopalakrishnan Meena, Antigoni Georgiadou, Gottiparthi Kalyan, Michael Sandoval, Antonio Coello Perez, Paul Lin, In-Saeng Suh and Seongmin Kim

690 — Simon's Period Finding on a Quantum Annealer — Reece Robertson, Emery Doucet, Zakaria Mzaouali, Krzysztof Domino, Bartłomiej Gardas and Sebastian Deffner

TP44::QALG — Adaptive Strategies for Variational Quantum Optimization

Wednesday, 3 September, 13:00 – 14:30

301 — Qracle: A Graph-Neural-Network-based Parameter Initializer for Variational Quantum Eigensolvers — Chi Zhang, Lei Jiang and Fan Chen

624 — An Adaptive Weighted QITE-VQE Algorithm for Combinatorial Optimization Problems — Ningyi Xie, Xinwei Lee, Tiejun Chen, Yoshiyuki Saito, Nobuyoshi Asai and Dongsheng Cai

193 — Quantum-Guided Cluster Algorithms for Combinatorial Optimization — Peter J. Eder, Aron Kerschbaumer, Jernej Rudi Finžgar, Raimel A. Medina, Martin J. A. Schuetz, Helmut G. Katzgraber, Sarah Braun and Christian B. Mendl

TP45::QALG — Efficient Quantum State Preparation and Algorithmic Primitives

Wednesday, 3 September, 15:00 – 16:30

425 — Generic State Preparation with Sub-Linear Depth Circuits of Multi-Controlled NOT Gates — Vittorio Pagni, Gary Schmiedinghoff, Kevin Lively, Michael Epping and Michael Felderer

585 — Efficient Gaussian State Preparation in Quantum Circuits — Yichen Xie and Nadav Ben-Ami

614 — Hybrid and Quantum Graph Algorithms via Quantum Array Search: Applications to Hamiltonian Paths and Path Counting — Ethan Hunt, Hieu Nguyen, Dheeraj Tukugudam, Dillon Horton and Tu Nguyen

TP46::QNET — Quantum State Engr & Network Interoperability**Wednesday, 3 September, 10:00 – 11:30**

319 — A double selection entanglement distillation-based state estimator — Joshua Carlo Casapao, Ananda Gopal Maity, Naphan Benchasattabuse, Michal Hajdusek, Akihito Soeda, Rodney Van Meter and David Elkouss

480 — Quantum State Transfer between Heterogenous 3D QPUs — Joseph Yaker, Silvia Zorzetti, Alessandro Reineri, Anna Grasselino, Yu-Chiu Chao, Tanay Roy, Alexander Romanenko, Jens Koch and Yao Lu

670 — From Physical to Logical: Graph-State-Based Connectivity in Quantum Networks — Mateo Blanco Rodríguez, Manuel Fernández Veiga, Ana Fernández Vilas and Rebeca P. Díaz Redondo

TP47::QNET — Entanglement in Quantum Networks I**Wednesday, 3 September, 13:00 – 14:30**

272 — Entanglement-Efficient Distribution of Quantum Circuits over Large-Scale Quantum Networks — Felix Burt, Kuancheng Chen and Kin K. Leung

436 — Efficient Execution of Multiple Quantum Circuits over a Quantum Network — Ranjani G Sundaram, Yukun Yang and Himanshu Gupta

662 — Scalable Time-Tagged Data Acquisition for Entanglement Distribution in Quantum Networks — Abderrahim Amlou, Thomas Gerrits, Anouar Rahmouni, Amar Abane, Abdella Battou, Mheni Merzouki, Oliver Slattery, Ya-Shian Li-Baboud and Ahmed Lbath

TP48::QNET — Entanglement in Quantum Networks II**Wednesday, 3 September, 15:00 – 16:30**

697 — Hybrid repeaters with encoding for long distance entanglement distribution — Stav Haldar, Saikat Guha, Don Towsley and Filip Rozpedek

792 — Network-Integrated Decoding System for Real-Time Quantum Error Correction with Lattice Surgery — Namitha Liyanage, Yue Wu, Emmet Houghton and Lin Zhong

322 — Self-Adaptive Quantum and Classical Software Systems — Antonio Brogi, Schahram Dustdar, Michael Felderer, Hausi Muller, Juan Manuel Murillo, Ricardo Pérez-Castillo, Antonio Ruiz-Cortés, Shinobu Saito and Ulrike Stege

TP49::QSYS — Quantum Circuit Optimization**Wednesday, 3 September, 10:00 – 11:30**

311 — Local Optimization of Quantum Circuits — Jatin Arora, Mingkuan Xu, Sam Westrick, Pengyu Liu, Dantong Li, Yongshan Ding and Umut A. Acar

345 — Quantum Circuit Optimization for the Fault-Tolerance Era: Do We Have to Start from Scratch? — Tobias Forster, Nils Quetschlich and Robert Wille

419 — Predict and Conquer: Navigating Algorithm Trade-offs with Quantum Design Automation — Simon Thelen and Wolfgang Maurer

TP50::QTEM — Quantum Technologies Optimization

Wednesday, 3 September, 10:00 – 11:30

176 — Optimizing Compilation for Distributed Quantum Computing via Clustering and Annealing — Ruilin Zhou, Jinglei Cheng, Yuhang Gan, Junyu Liu and Chen Qian

318 — Towards Efficient Verification of Computation in Quantum Devices — Keren Li, Peng Yan, Hanru Jiang and Nengkun Yu

616 — Optimized GKP State for Bosonic Channel Sensing — Thinh Le, Jianqing Liu, Jiapeng Zhao and Eneet Kaur

TP51::QALG — Quantum Data Encoding and Signal Processing for Reliable Computation

Wednesday, 3 September, 10:00 – 11:30

459 — An Efficient Algorithm for Extending Uniform Superposition States for Time-Series Data — Carla Rieger, Michele Grossi, Gian Giacomo Guerreschi, Sofia Vallecorsa and Martin Werner

787 — Statistical Signal Processing for Quantum Error Mitigation — Kausthubh Chandramouli, Kelly Mae Allen, Christopher Mori, Dror Baron and Mário A. T. Figueiredo

644 — On the Correct Implementation of the Walsh Series Loader — Riccardo Pellini and Maurizio Ferrari Dacrema

TP52::QALG — Quantum Circuit Design for Modeling, Estimation, and Control

Wednesday, 3 September, 15:00 – 16:30

688 — Quantum Solution Framework for Finite-Horizon LQG Control via Block Encodings and QSVT — Nahid Binandeh Dehaghani, Rafal Wisniewski and A. Pedro Aguiar

271 — Quantum Circuit Design for Decoded Quantum Interferometry — Natchapol Patamawisut, Naphan Benchasattabuse, Michal Hajdušek and Rodney Van Meter

199 — Optimizing State Preparation for Variational Quantum Regression on NISQ Hardware — Frans Perkkola, Ilmo Salmenperä, Arianne Meijer-van de Griend, C.-C. Joseph Wang, Ryan S. Bennink and Jukka K. Nurminen

TP53::QML_ — Structure-Aware QML for Physical and Graph-Based Systems

Wednesday, 3 September, 15:00 – 16:30

491 — Autoregressive pairwise Graphical Models efficiently find ground state representations of stoquastic Hamiltonians — Yuchen Pang, Abhijith Jayakumar, Evan McKinney, Carleton Coffrin, Marc Vuffray and Andrey Lokhov

374 — Solving graph problems using permutation-invariant quantum machine learning — Maximilian Balthasar Mansky, Tobias Rohe, Gerhard Stenzel, Leo Sünkel, Alejandro Bravo de la Serna, Santiago Londoño Castillo, Gautham Sathish, Dimitra Nikolaidou, Dmytro Bondarenko, Linus Menzel and Claudia Linnhoff-Popien

481 — One, Two, Three: One Empirical Evaluation of a Two-Copy Shadow Tomography Scheme with Triple Efficiency — Viet Thuong Tran and Richard Kueng

TP54::QSYS — Quantum Error Correction I
Thursday, 4 September, 10:00 – 11:30

182 — Enhancing the Clique Local Decoder to Correct Length-2 Space Errors in the Surface Code — Zikang Jia, Shravan Veerapaneni and Gokul Subramanian Ravi

210 — ConiQ: Enabling Concatenated Quantum Error Correction on Neutral Atom Arrays — Pengyu Liu, Mingkuan Xu, Hengyun Zhou, Hanrui Wang, Umut A. Acar and Yunong Shi

489 — Surface Code Error Correction with Crosstalk Noise — Zeyuan Zhou, Andrew Ji and Yongshan Ding

TP55::QSYS — Quantum Error Correction II
Thursday, 4 September, 13:00 – 14:30

523 — Moveless: Minimizing QEC Overhead on QCCDs via Versatile Execution and Low Excess Shuttling — Sahil Khan, Suhas Vittal, Kenneth Brown and Jonathan Baker

349 — Distributed quantum error correction based on hyperbolic Floquet codes — Evan Sutcliffe, Bhargavi Jonnadula, Claire Le Gall, Alexandra E. Moylett and Coral M. Westoby

812 — Erasure Minesweeper: exploring hybrid-erasure surface code architectures for efficient quantum error correction — Jason D. Chadwick, Mariesa H. Teo, Joshua Viszlai, Willers Yang and Frederic T. Chong

TP56::QSYS — Quantum Error Correction III
Thursday, 4 September, 15:00 – 16:30

190 — Locality-aware Pauli-based computation for local magic state preparation — Yutaka Hirano and Keisuke Fujii

183 — Improving Hardware Requirements for Fault-Tolerant Quantum Computing by Optimizing Error Budget Distributions — Tobias Forster, Nils Quetschlich, Mathias Soeken and Robert Wille

206 — Matching Generalized-Bicycle Codes to Neutral Atoms for Low-Overhead Fault-Tolerance — Joshua Viszlai, Willers Yang, Sophia Fuhui Lin, Junyu Liu, Natalia Nottingham, Jonathan Baker and Fred Chong

TP57::QNET — Routing in Quantum Networks
Thursday, 4 September, 10:00 – 11:30

401 — Entanglement routing over networks with time multiplexed repeaters — Emily Van Milligen, Eliana Jacobson, Ashlesha Patil, Gayane Vardoyan, Don Towsley and Saikat Guha

708 — Statistical Modeling and Latency Optimization for Entanglement Routing in Quantum Networks — Jiyao Liu and Yu Wang

774 — Beyond Traditional Quantum Routing — Siyi Chen, Angela Sara Cacciapuoti and Marcello Caleffi

TP58::QNET — Quantum Network Design

Thursday, 4 September, 13:00 – 14:30

107 — ShaNQar: Simulator of Network Quantique — Anand Choudhary and Ajay Wasan

189 — On Designing A Quantum Network Topology: Model, Methodology, and Algorithm — Yuhang Gan, Ruilin Zhou, Shiyi Ling, Yi Liu and Chen Qian

359 — A tailored fidelity estimation and purification method for bootstrapping a quantum network — Takafumi Oka, Michal Hajdusek, Shota Nagayama and Rodney Van Meter

TP59::QNET — Quantum Paths

Thursday, 4 September, 15:00 – 16:30

284 — On Selecting Paths for End-to-End Entanglement Creation in Quantum Networks — Anoosha Fayyaz, Prashant Krishnamurthy, Kaushik Seshadreesan, David Tipper and Amy Babay

663 — Towards Optimal Orders for Entanglement Swapping in Path Graphs: A Greedy Approach — Van Sy Mai, Abderrahim Amlou, Amar Abane and Abdella Battou

755 — Quantum Paths: a Quantum Walk approach — Claudio Pellitteri, Marcello Caleffi and Angela Sara Cacciapuoti

TP60::QPHO — Integrated Photonic Technologies for Scalable Quantum Systems

Thursday, 4 September, 10:00 – 11:30

970 — Sub nanosecond time-gated superconducting nanowire single-photon detectors (SNSPDs) — Antonio Guardiani, Katyayani Seal, Andreas Fognini, Lieuwe Locht and Henri Vlot

395 — Quantum Integrated Photonic Network-on-Chip for Manycore Architectures — Zach Takacs, Harsha Chenji and Avinash Karanth

744 — Nonclassical Light Generation in Quantum Dot Microlasers: Toward Integrated Quantum Photonic Sources — Frederic Grillot, Weng Chow, Galan Moody and Chen Shang

TP61::QPHO — Quantum Advantage and Benchmarking in Optical and Photonic Systems**Thursday, 4 September, 13:00 – 14:30**

836 — Infinite-fold Quantum Advantage in Classical Optical Correlation Sensing — Janis Nötzel and Pere Munar-Vallespir

855 — Gauging practical computational advantage using a classical, threshold-based Gaussian boson sampler — Sarvesh Raghuraman, Aditya Patwardhan and Brian La Cour

925 — A Digital Twin for Multimode Quantum Entanglement Transmission in Noisy and Lossy Optical Channels — Alireza Alipour and Jonathan Habif

TP62::QPHO — Advances in Photonic Quantum Computing**Thursday, 4 September, 15:00 – 16:30**

860 — On the Mitigability of Probabilistic Photonics CNOT Failures of Variational Quantum Classifiers — Muhammad Hosam, Muhammad Ahmed, Norhan Abd Elgawad, Tamer Abuefadel and Ahmed El-Mahdy

337 — Distributed Quantum Neural Networks on Distributed Photonic Quantum Computing — Kuan-Cheng Chen, Chen-Yu Liu, Felix Burt, Shang Yu and Kin Leung

846 — Towards Synalgebraic Photonic Quantum Computing — Anderson Beraldo de Araujo

TP63::QALG — Quantum Algorithms & Benchmarking for Scientific & Optimization Workflows**Thursday, 4 September, 15:00 – 16:30**

232 — Adaptive mesh refinement quantum algorithm for Maxwell's equations — Elise Fressart, Michel Nowak and Nicole Spillane

702 — Optimal Scaling Quantum Interior Point Method for Linear Optimization — Mohammadhossein Mohammadisiahroudi, Zeguan Wu, Pouya Sampourmahani, Jun-Kai You and Tamás Terlaky

497 — Benchmarking Fidelity Metrics of Quantum Computers — Shubdeep Mohapatra, Hrushikesh Pramod Patil, Ji Liu and Huiyang Zhou

TP64::QALG — Foundations of Quantum Operations**Thursday, 4 September, 10:00 – 11:30**

95 — QuOp: A Quantum Operator Representation for Nodes — Andrew Vlasic and Salvador Aguinaga

187 — Quantum Arithmetic Algorithms: Implementation, Resource Estimation, and Comparison — Dmytro Fedoriaka, Brian Goldsmith and Yingrong Chen

263 — Learning to Program Quantum Measurements for Machine Learning — Samuel Yen-Chi Chen, Huan-Hsin Tseng, Hsin-Yi Lin and Shinjae Yoo

TP65::QSYS — Quantum Circuit Compilation I

Thursday, 4 September, 10:00 – 11:30

155 — BVQC: A Backdoor-style Watermarking Scheme for Variational Quantum Circuits — Cheng Chu, Lei Jiang and Fan Chen

513 — Task Division of Quantum Circuits using ZX-calculus — Navnil Choudhury, Ameya Shirish Bhawe and Kanad Basu

423 — Physically-Aware Qubit Reuse for Optimized Quantum Circuit Compilation Using Machine Learning — Gopika Kizhuvettil, Georgios Ioannou, Siyuan Niu and Samah Saeed

TP66::QSYS — Quantum Circuit Compilation II

Thursday, 4 September, 15:00 – 16:30

160 — Compilation strategies for quantum network programs using Qoala — Samuel Oslovich, Bart van der Vecht and Stephanie Wehner

273 — Improving Quantum Compilation via Circuit Partitioning and Floorplanning — Giacomo Lancellotti, Giovanni Agosta, Alessandro Barenghi and Gerardo Pelosi

645 — TeleSABRE: Heuristic Layout Synthesis in Multi-Core Quantum Systems with Teleport Interconnect — Enrico Russo, Elio Vinciguerra, Maurizio Palesi, Davide Patti, Giuseppe Ascia and Vincenzo Catania

TP67::QML_ — AI-Driven Generation and Optimization of Quantum Circuits

Friday, 5 September, 13:00 – 14:30

321 — Agent-Q: Fine-Tuning Large Language Models for Quantum Circuit Generation and Optimization — Linus Jern, Valter Uotila, Cong Yu and Bo Zhao

604 — Leveraging Diffusion Models for Parameterized Quantum Circuit Generation — Daniel Barta, Darya Martyniuk, Johannes Jung and Adrian Paschke

180 — CleanQRL: Lightweight Single-file Implementations of Quantum Reinforcement Learning Algorithms — Georg Kruse, Rodrigo Coelho, Andreas Rosskopf, Robert Wille and Jeanette Miriam Lorenz

TP68::QML_ — Secure and Feedback-Driven QML

Friday, 5 September, 15:00 – 16:30

257 — Feedback Connections in Quantum Reservoir Computing with Mid-Circuit Measurements — Jakob Murauer, Rajiv Krishnakumar, Sabine Törnøw and Michaela Geierhos

291 — Entangled Threats: A Unified Kill Chain Model for Quantum Machine Learning Security — Pascal Debus, Kilian Tscharke, Maximilian Wendlinger, Daniel Herr, Cedric Brüggmann, Daniel Ohl de Mello, Juris Ulmanis, Alexander Erhard, Fabian Petsch and Arthur Schmidt

771 — Quantum Machine Learning for Intrusion Detection on Noisy Quantum Computers — Franco Cirillo and Christian Esposito

TP69::QAPP — Quantum Computing for Biological systems and Biomedical Data
Friday, 5 September, 10:00 – 11:30

508 — Towards secondary structure prediction of longer mRNA sequences using a quantum-centric optimization scheme — Vaibhaw Kumar, Dimitris Alevras, Mihir Metkar, Eline Welling, Chris Cade, Ido Niesen, Triet Friedhoff, Jae-Eun Park, Saurabh Shivpuje, Mariana LaDue, Wade Davis and Alexey Galda

511 — Co-optimization of codon usage and mRNA secondary structure using quantum computing — Dimitris Alevras, Mihir Metkar, Triet Friedhoff, Jae-Eun Park, Mariana LaDue, Vaibhaw Kumar, Wade Davis and Alexey Galda

269 — Resting-state fMRI Analysis using Quantum Time-series Transformer — Junghoon Justin Park, Jungwoo Seo, Sangyoon Bae, Samuel Yen-Chi Chen, Huan-Hsin Tseng, Jiook Cha and Shinjae Yoo

TP70::QAPP — Quantum Techniques for Classical Computing and Language Processing
Friday, 5 September, 13:00 – 14:30

463 — On the Potential of Quantum Computing in Classical Program Analysis — Yicheng Guang, Pietro Zanotta, Kai Zhou, Yueqi Chen and Ramin Ayanzadeh

369 — Quantum Language Model Fine Tuning — Sang Kim, Jonathan Mei, Claudio Grotto, Masako Yamada and Martin Roetteler

323 — SQL2Circuits: Estimating Cardinalities, Execution Times, and Costs for SQL Queries with Quantum Natural Language Processing — Valter Uotila

TP71::QAPP — Quantum-Inspired Models for Learning, Perception, and Cognition
Friday, 5 September, 15:00 – 16:30

675 — k-Contextuality as a Heuristic for Memory Separations in Learning — Mariesa Teo, Willers Yang, James Sud, Teague Tomesh, Frederic Chong and Eric Anschuetz

791 — Towards Hybrid Spatial Structure Rendering Models Using Quantum Orthographic Ray Casting — Zirui Zhang

471 — Quantum Integration of Spiking Leaky Integrate-and-Fire Neurons and Variational Circuits for Enhanced Multiclass Classification — Mandeep Saggi, Amandeep Bhatia and Sabre Kais

TP72::QML_ — Architectures and Preprocessing for Scalable Scientific QML
Friday, 5 September, 10:00 – 11:30

775 — Partitioned Hybrid Quantum Fourier Neural Operators for Scientific Quantum Machine Learning — Paolo Marcandelli, Yuanchun He, Stefano Mariani, Martina Siena and Stefano Markidis

388 — Influence of Data Dimensionality Reduction Methods on the Effectivity of Quantum Machine Learning Models — Aakash Ravindra Shinde and Jukka K. Nurminen

727 — Generative Modeling of Measurement Data for Quantum Parameter Adaptation — Chen-Yu Liu, Samuel Yen-Chi Chen, Kuan-Cheng Chen, Wei-Jia Huang, Wei-Hao Huang and Yen-Jui Chang

TP73::QML_ — Improving Variational Learning

Friday, 5 September, 13:00 – 14:30

234 — Can Classical Initialization Help Variational Quantum Circuits Escape the Barren Plateau? — Yifeng Peng, Xinyi Li, Zhemin Zhang, Samuel Yen-Chi Chen, Zhiding Liang and Ying Wang

607 — Breaking Through Barren Plateaus: Reinforcement Learning Initializations for Deep Variational Quantum Circuits — Yifeng Peng, Xinyi Li, Zhemin Zhang, Samuel Yen-Chi Chen, Zhiding Liang and Ying Wang

347 — Variational Quantum Circuits in Offline Contextual Bandit Problems — Lukas Schulte, Daniel Hein, Steffen Udluft and Thomas A. Runkler

TP74::QML_ — Quantum Architecture Search

Friday, 5 September, 15:00 – 16:30

598 — QAS-QTNs: Curriculum Reinforcement Learning-Driven Quantum Architecture Search for Quantum Tensor Networks — Siddhant Dutta, Nouhaila Innan, Sadok Ben Yahia and Muhammad Shafique

684 — Neural Architecture Search for Quantum Autoencoders — Hibah Agha, Samuel Yen-Chi Chen, Huan-Hsin Tseng and Shinjae Yoo

695 — Q-RLONAS: Towards Efficient Quantum Neural Architecture Search — Seok Bin Son and Soohyun Park

TP76::QAPP — Simulating the Complex: Quantum Methods for Fields, Flows, and Forecasting

Friday, 5 September, 13:00 – 14:30

407 — Algorithmic Advances towards a Realizable Quantum Lattice Boltzmann Method — Apurva Tiwari, Jason Iaconis, Jezer Jojo, Sayonee Ray, Martin Roetteler, Chris Hill and Jay Pathak

680 — Flowing Through Hilbert Space: Quantum-Enhanced Generative Models for Lattice Field Theory — Jehu Martinez and Andrea Delgado

368 — Quantum Autoencoder for Multivariate Times Series Anomaly Detection — Kilian Tscharke, Maximilian Wendlinger, Afrae Ahouzi, Pallavi Bhardwaj, Kaweh Amoi-Taleghani, Michael Schrödl-Baumann and Pascal Debus

TP77::QAPP — Scalable Quantum Error Correction**Friday, 5 September, 15:00 – 16:30**

473 — Distributed Realization of Color Codes for Quantum Error Correction — Nitish Chandra, David Tipper, Reza Nejabati, Eneet Kaur and Kaushik Seshadreesan

773 — Disparity Surface Code: Optimizing Error Correction in Quantum Networks — Tianjie Hu, Jindi Wu and Qun Li

220 — Fault Tolerant Decoding of QLDPC-GKP Codes with Circuit Level Soft Information — Shantom K. Borah, Asit K. Pradhan, Nithin Raveendran, Michele Pacenti and Bane Vasić

TP78::QTEM — Quantum Control I**Friday, 5 September, 10:00 – 11:30**

696 — High output SoC-based ion shuttling waveform generator using cubic splines — Tiamike Dudley, Jim Plusquellic, Joshua Goldberg, Dan Stick and Dan Lobser

123 — Multiplexed Control at Scale for Electrode Arrays in Trapped-Ion Quantum Processors — Ryutaro Ohira, Shinichi Morisaka, Ippei Nakamura, Atsushi Noguchi and Takefumi Miyoshi

292 — Real-time DCZ Gate Control by Bayesian Optimization with Forgetting — Yudai Sato and Takayuki Kawahara

TP79::QTEM — Quantum Control II**Friday, 5 September, 13:00 – 14:30**

328 — Iterative Linear Quadratic Regulator for Quantum Optimal Control — Dirk Heimann, Felix Wiebe, Tahereh Abad, Elie Mounzer, Tangyou Huang, Frank Kirchner and Shivesh Kumar

818 — A Modular ATCA-Based Approach Towards Scalable Multi-Qubit Systems — Lukas Scheller, Marvin Fuchs, Robert Gartmann, Timo Muscheid, Roger Heil, Christian Roth, Oliver Sander, Mario Schloesser and Luis Ardila-Perez

631 — Cryo-CMOS control modeling for fluxonium qubits — Vladimir Pesic, Oliver Smedt, Andrew Wright, Raymond Mencia, Vladimir Manucharyan and Edoardo Charbon

TP80::QTEM — Co-simulation and Analysis for Superconducting Platforms**Friday, 5 September, 15:00 – 16:30**

281 — Nonlinear co-simulation for designing kinetic inductance parametric amplifiers — Likai Yang, Yufeng Wu, Chaofan Wang, Mingrui Xu, Hong Tang, Mohamed Hassan and Eric Holland

679 — Parametric Stability Analysis for Circuit Quantum Electrodynamical Hardware — Maria Gabriela Boada, Andrea Delgado and Jose Morales

785 — Analysis of a 3D Integrated Superconducting Quantum Chip Structure — James Saslow and Hiu-Yung Wong

TP81::QSYS — Tools for Reliable Quantum Computing

Friday, 5 September, 10:00 – 11:30

236 — Rule-based hardware-configurable static analysis for quantum programs — Yi-Ting Chen, Lauren Capelluto, Ryan Shaffer and Jeffrey Heckey

327 — Automatically Refining Assertions for Efficient Debugging of Quantum Programs — Damian Rovara, Lukas Burgholzer and Robert Wille

822 — Secure Quantum Circuit Compilation Methodology for Untrusted Compilers — Subrata Das and Swaroop Ghosh

TP82::QSYS — Software Infrastructure for Scalable Quantum Execution

Friday, 5 September, 13:00 – 14:30

191 — Online Job Scheduler for Fault-tolerant Quantum Multiprogramming — Ryo Wakizaka, Shin Nishio, Daisuke Sakuma, Yosuke Ueno and Yasunari Suzuki

332 — Quantum Island Mapping: Optimizing Multi-Circuit Execution in Quantum Processors — Jaime Alvarado-Valiente, Javier Romero-Álvarez, Jorge Casco-Seco, Enrique Moguel and Jose Garcia-Alonso

315 — A Practical Open-Source Software Stack for a Cloud-Based Quantum Computing System — Norihiro Kakuko, Shun Gokita, Naoyuki Masumoto, Keita Matsumoto, Kosuke Miyaji, Takafumi Miyanaga, Toshio Mori, Haruki Nakayama, Keita Sasada, Yasuhito Takamiya, Satoyuki Tsukano, Ryo Uchida and Masaomi Yamaguchi

TP83::QML_ — Quantum Kernels and Structured Circuits for Supervised Learning

Friday, 5 September, 10:00 – 11:30

759 — Support Vector Classification and Regression using an Adaptive Quantum-Classical Dual Kernel — Dhana Phassadawongse and Stephen John Turner

150 — Quantum Kernel Anomaly Detection Using AR-Derived Features from Non-Contact Acoustic Monitoring for Smart Manufacturing — Takao Tomono and Kazuya Tsujimura

153 — Fourier Analysis of Variational Quantum Circuits for Supervised Learning — Marco Wiedmann, Maniraman Periyasamy and Daniel David Scherer

TP84::QML_ — Data-Efficient and Uncertainty-Aware QML

Friday, 5 September, 13:00 – 14:30

475 — Quantum Machine Learning for minimal omics datasets with large feature space using embeddings and feature selection techniques — Emre Sahin, Matthew Madgwick, James Strudwick, Vivek Sehra, Oscar Wallis, Hamed Javidi Mostafapour Boshroyeh, Anna Paola Carrieri, Laxmi Parida, Lara Jehi, Stefano Mensa and Filippo Utró

595 — Old Rules in a New Game: Mapping Uncertainty Quantification to Quantum Machine Learning — Maximilian Wendlinger, Kilian Tschärke and Pascal Debus

252 — Quantum Reupload Units: A Scalable and Expressive Approach for Time Series Learning — Léa Cassé, Sabarikirishwaran Ponnambalam, Bernhard Pfahringer and Albert Bifet

TP85::QSYS — Quantum Programming Languages
Friday, 5 September, 10:00 – 11:30

7 — Qwerty: A Basis-Oriented Quantum Programming Language — Austin J. Adams, Sharjeel Khan, Arjun S. Bhamra, Ryan R. Abusaada, Travis S. Humble, Jeffrey S. Young and Thomas M. Conte

544 — Productive Quantum Programming Needs Better Abstract Machines — Santiago Núñez-Corrales, Olivia Di

TP86::QSYS — Quantum Error Mitigation
Friday, 5 September, 13:00 – 14:30

270 — Distinguishing Quantum Software Bugs from Hardware Noise: A Statistical Approach — Devraj D, Ahmik Virani, Anirudh Suresh, Lei Zhang and Mvprao

476 — Q-Cluster: Quantum Error Mitigation Through Noise-Aware Unsupervised Learning — Hrushikesh Pramod Patil, Dror Baron and Huiyang Zhou

536 — Optimal Partitioning for Noisy Phase Estimation with the Greenberger-Horne-Zeilinger (GHZ) States — Allen Zang, Tian-Xing Zheng, Peter C. Maurer, Frederic T. Chong, Martin Suchara and Tian Zhong

TP87::QNET — Quantum Networking Protocols I
Friday, 5 September, 10:00 – 11:30

172 — The Proportional Fair Scheduler in Wavelength-Multiplexed Quantum Networks — Sanidhay Bhambay, Siddarth Koduru Joshi, Thirupathaiah Vasantam and Neil Walton

226 — Automatic Configuration Protocols for Optical Quantum Networks — Amin Taherkhani, Andrew Todd, Kentaro Teramoto, Shota Nagayama and Rodney Van Meter

353 — Simulation of Quantum Transduction Strategies for Quantum Networks — Laura D'Avossa, Caitao Zhan, Joaquin Chung, Rajkumar Kettimuthu, Angela Sara Cacciapuoti and Marcello Caleffi

TP88::QNET — Quantum Networking Protocols II
Friday, 5 September, 13:00 – 14:30

455 — Accurate and Scalable Simulation of Cavity-Based Networks in Modular Quantum Architectures — Sahar Ben Rached, Zezhou Sun, Guilu Long, Santiago Rodrigo Munoz, Carmen Garcia Almudever, Eduard Alarcon and Sergi Abadal

689 — Flexible Superconducting Interconnects Realized by Electrodeposition of Rhenium on an Aerosol Jet Printed Seed Layer — Jacob Manzi, Lok-Kun Tsui, Yongkun Sui, Kamal Ahammed, Zhefan Ma, Qiang Huang and Judith Lavin

492 — A Two-level Control Framework for Quantum Networks — Se-young Yu, Elia Perego, Justin Philips, You-Wei Cheah, Prathwiraj Umesh, Guangqi Gao, Jiarui Liu, Ezra Kissel, Michael Bregar, Ke Sun, Qiming Wu, Venkata Ramana Raju Valivarthi, Erhan Saglamyurek, Wenji Wu, Maria Spiropulu, Hartmut Häeffner and Inder Monga

TP89::QALG — Quantum Algorithms
Tuesday, 2 September, 15:00 – 16:30

426 — Efficient QAOA Architecture for Solving Multi-Constrained Optimization Problems — David Bucher, Maximilian Janetschek, Daniel Porawski, Jonas Stein, Corey O'Meara, Giorgio Cortiana and Claudia Linnhoff-Popien

178 — Autocallable Options Pricing with Integration-Based Exponential Amplitude Loading — Francesca Cibrario, Ron Cohen, Emanuele Dri, Christian Mattia, Or Samimi Golan, Tamuz Danzig, Giacomo Ranieri, Hanan Rosemarin, Davide Corbelleto, Amir Naveh and Bartolomeo Montrucchio

512 — QAOA-GPT: Efficient Generation of Adaptive and Regular Quantum Approximate Optimization Algorithm Circuits — Ilya Tyagin, Marwa Farag, Karunya Shirali, Kyle Sherbert, Yuri Alexeev and Ilya Saфро

TP90::SYS — Quantum Systems
Tuesday, 2 September, 10:00 – 11:30

780 — Exploring Beamforming Solutions using Recursive Quantum Approximate Optimization Algorithm — Bidisha Dhara, Monika Agrawal and Sumantra Dutta Roy

754 — Polynomial-time Extraction of Entanglement Resources — Siyi Chen, Angela Sara Cacciapuoti and Marcello Caleffi

TP91::SYS — Quantum Systems
Tuesday, 2 September, 13:00 – 14:30

583 — A Game-Theoretic Quantum Algorithm for Solving Magic Squares — Sarah Chehade, Andrea Delgado and Elaine Wong

651 — Compiler design for hardware specific decomposition optimizations, tailored to diamond NV centers — Folkert de Ronde, Stephan Wong and Sebastian Feld

560 — Quantum Adiabatic Generation of Human-Like Passwords — Sascha Mücke, Raoul Heese, Thore Gerlach, David Biesner, Loong Kuan Lee and Nico Piatkowski